

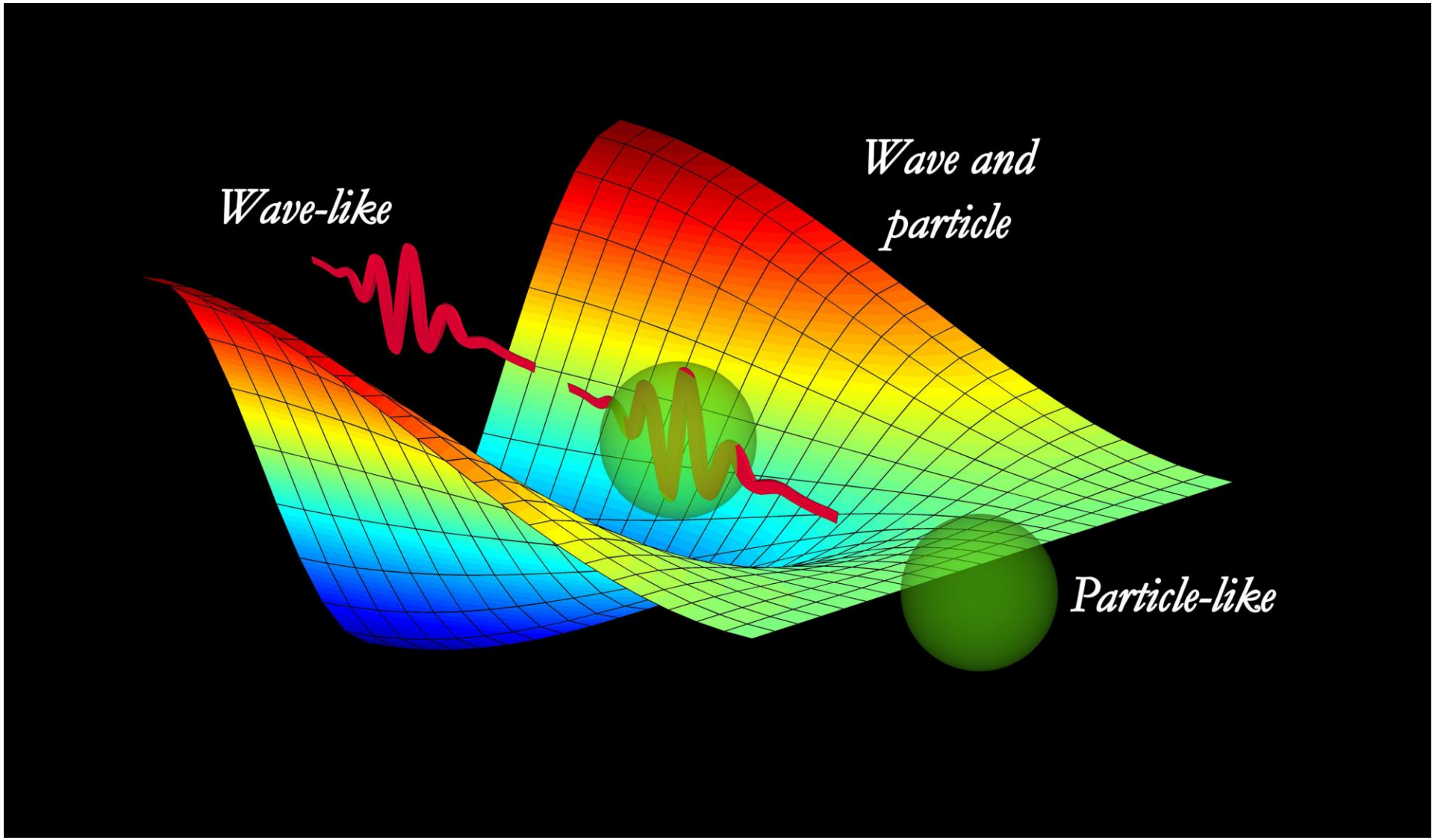
Rino R. Mukti
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Week 9

KI1101 General Chemistry 1

Atomic Structure

Duality of Light as Wave and Particle



Significant Experiments for Atomic Theory

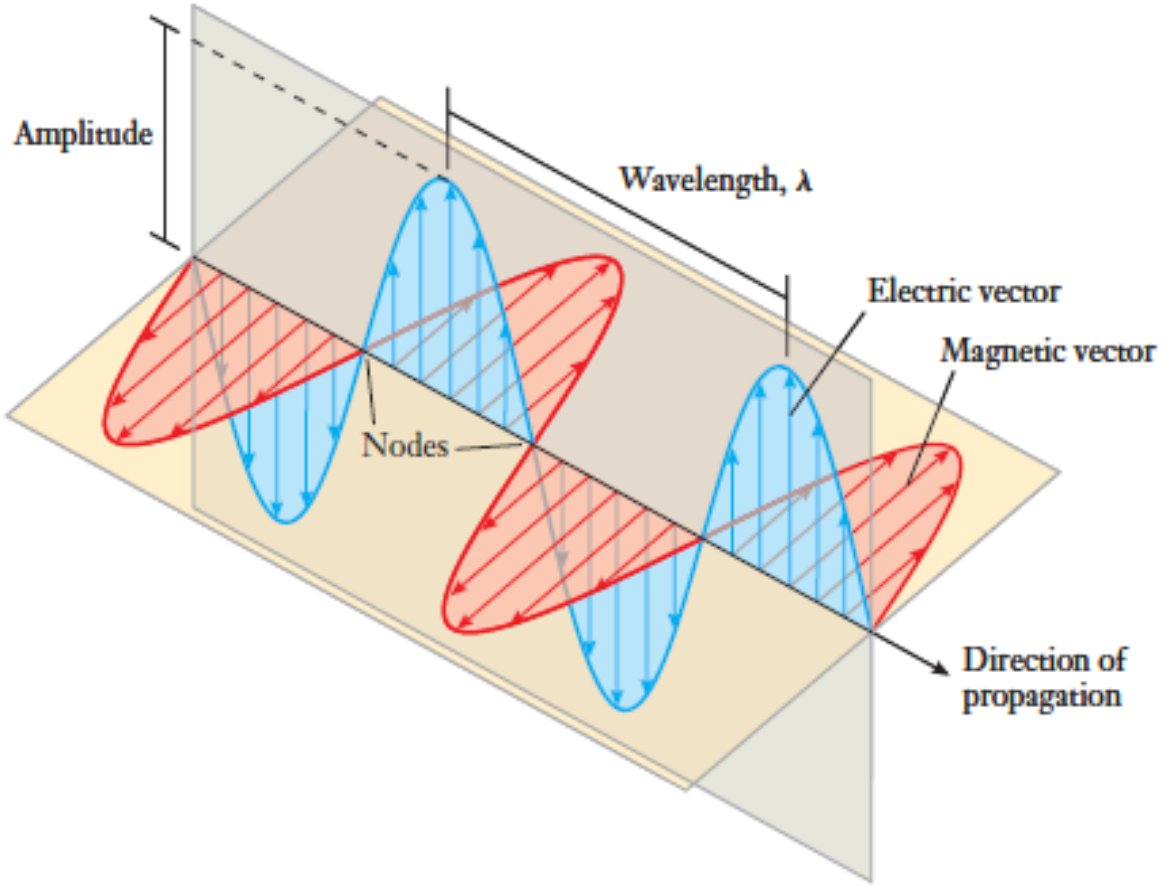
Late 1800's:

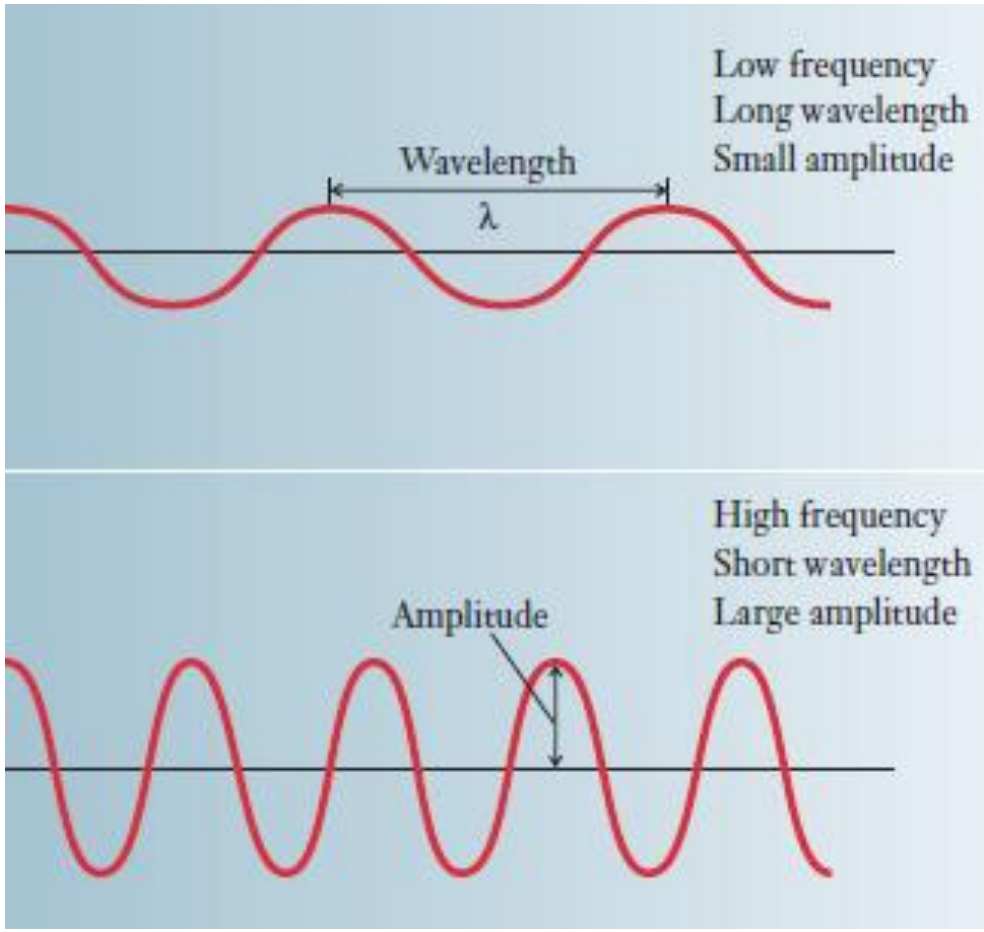
- Classical physics incapable of describing atoms and molecules
- Matter and energy believed to be distinct
  - **Matter:** made up of particles
  - **Energy:** light waves

Beginning of 1900's:

- Several experiments proved above ideas incorrect
- Experiments showed that electrons acted like:
  - Tiny charged particles in some experiments
  - Waves in other experiments

Light as Wave and Electromagnetic Radiation





Wave Properties

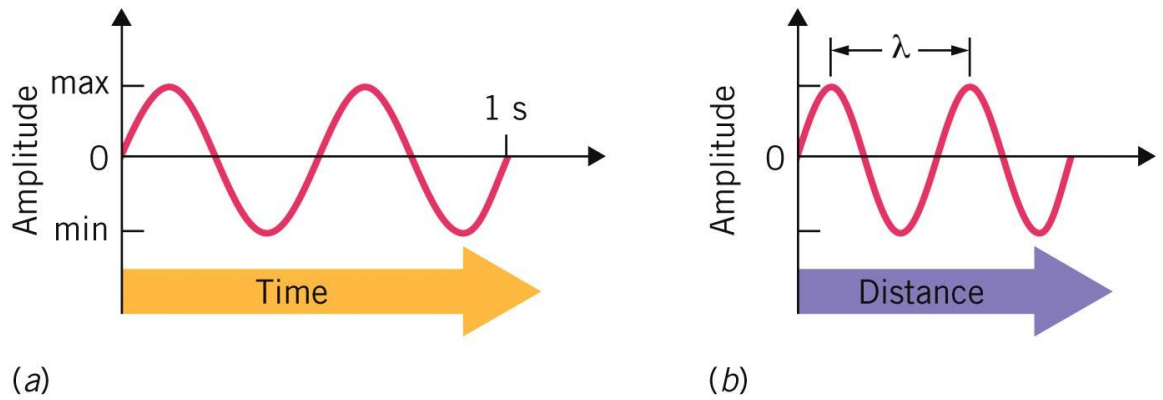
Wavelength (λ)

- Distance between two successive peaks or troughs
- Unit = **meter**

Frequency (ν)

- number of waves per second that pass a given point in space
- Unit = **Hertz** (Hz) = cycles/sec = 1/sec = s^{-1}

Related by $\lambda \times \nu = c$



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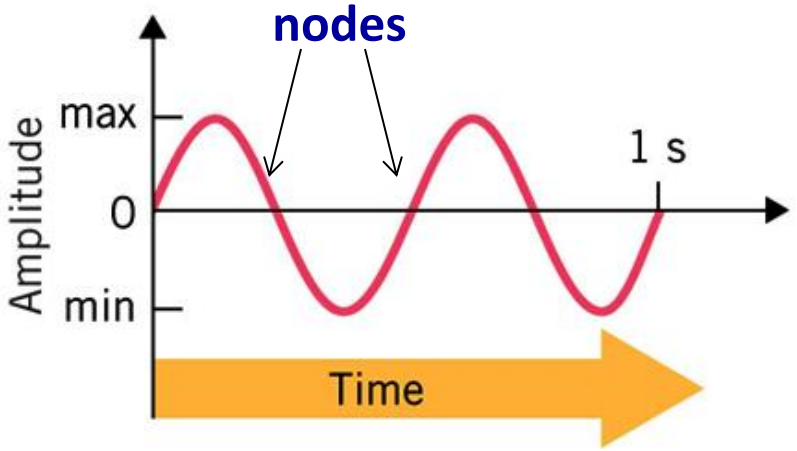
Wave Properties

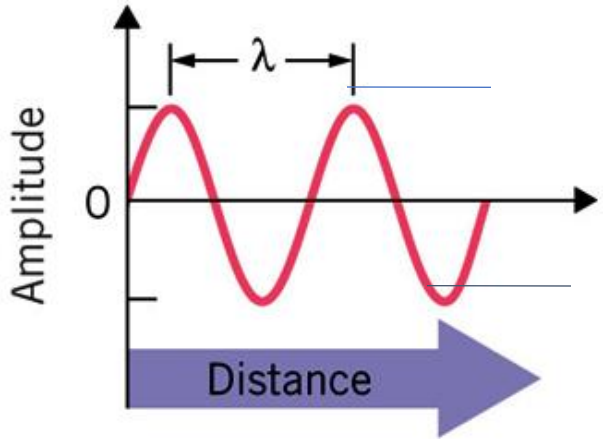
Amplitude

- Intensity of wave
- Maximum and minimum height
- Varies with time as travels through space

Nodes

- Points of zero amplitude
- Place where wave goes though axis
- Distance between nodes is always same

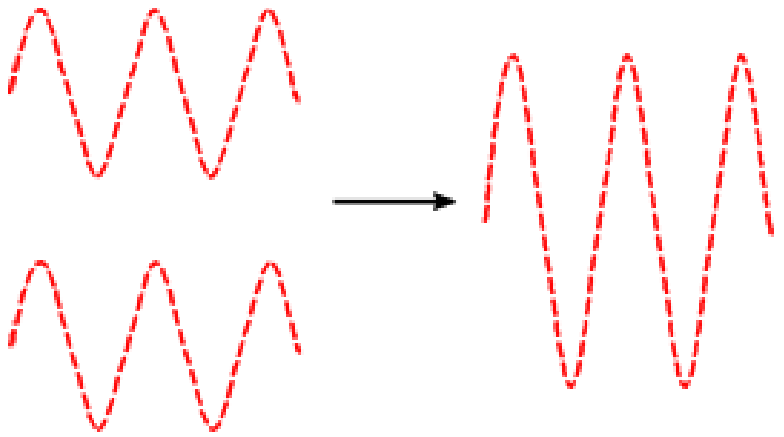




Wave Interference

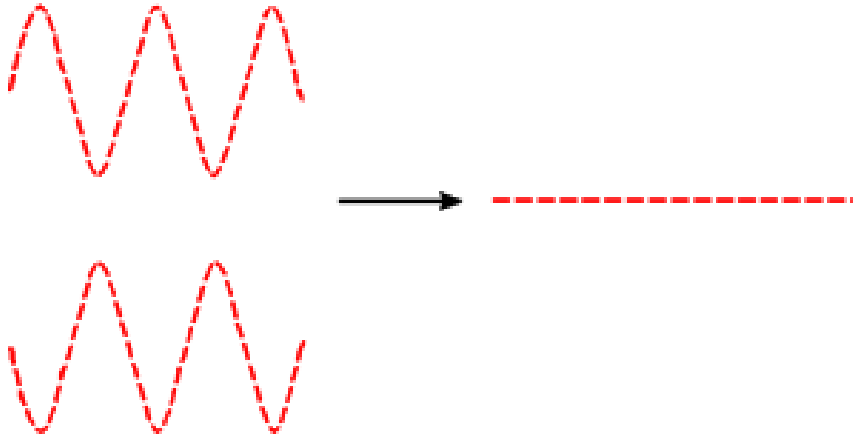
Constructive interference

- Waves “in-phase” lead to greater amplitude
- Add

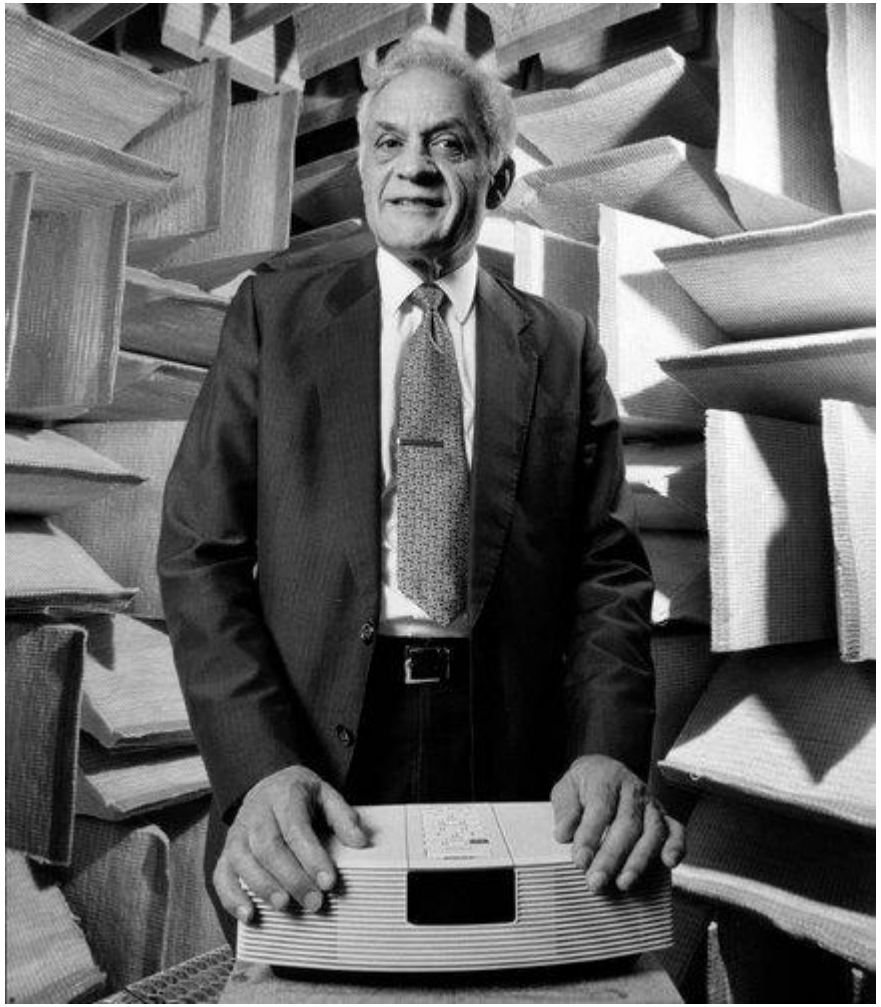


Destructive interference

- Waves “out-of-phase” lead to lower amplitude
- Cancel out




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Electromagnetic Radiation

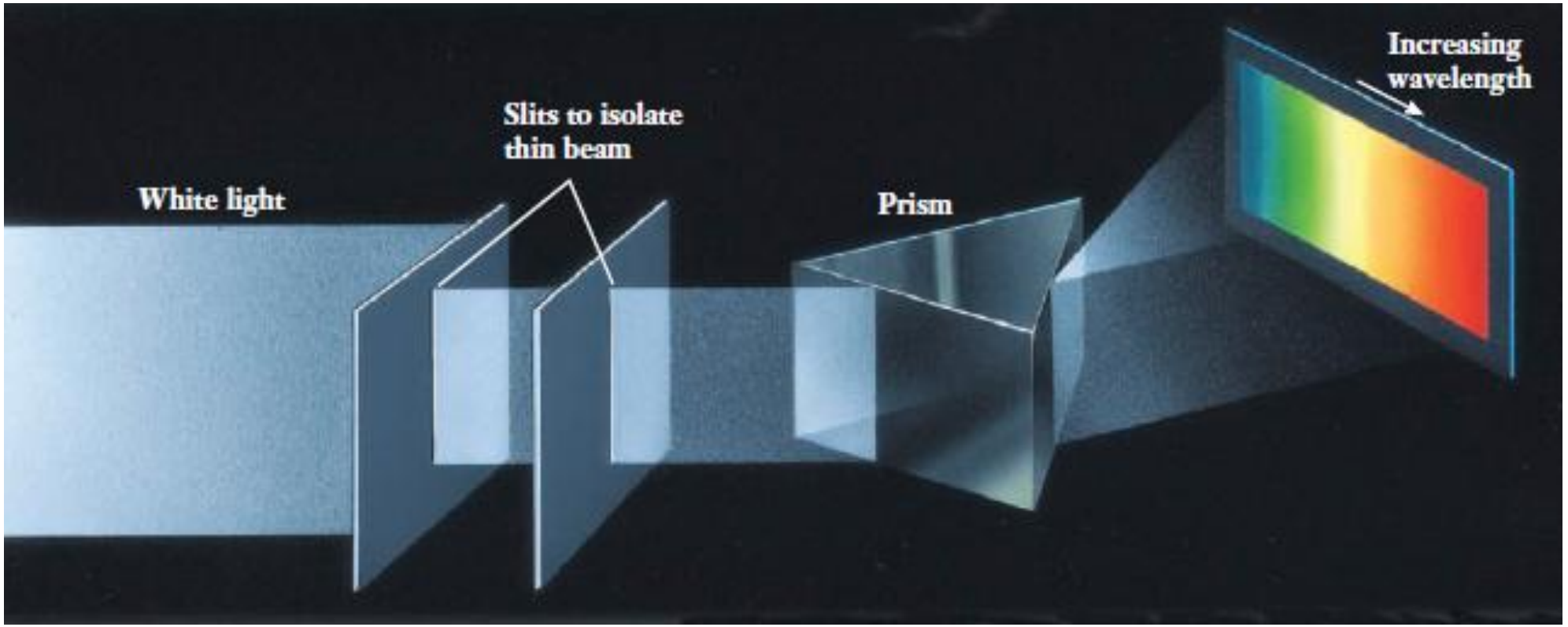
Electromagnetic Radiation

- Light energy or wave
- Travels through space at speed of light in vacuum
- ***c* = speed of light** = 2.9979×10^8 m/s
- Successive series of these waves or oscillations

Waves or **Oscillations**

- Systematic fluctuations in intensities of electrical and magnetic forces
- Varies rhythmically with time
- Exhibit wide range of energy

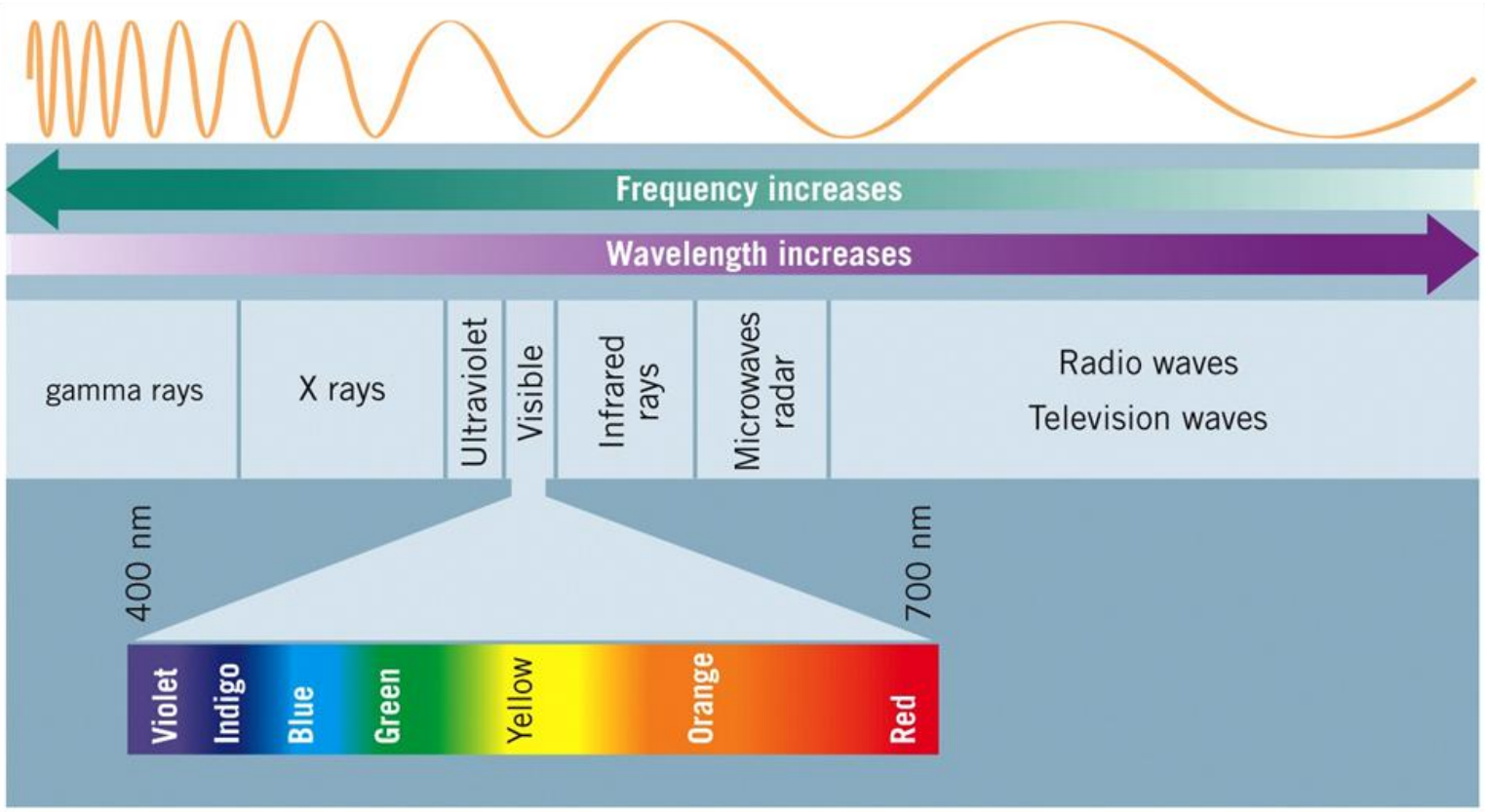
Visible Light Spectrum



Electromagnetic Spectrum

high energy, short waves

low energy, long waves

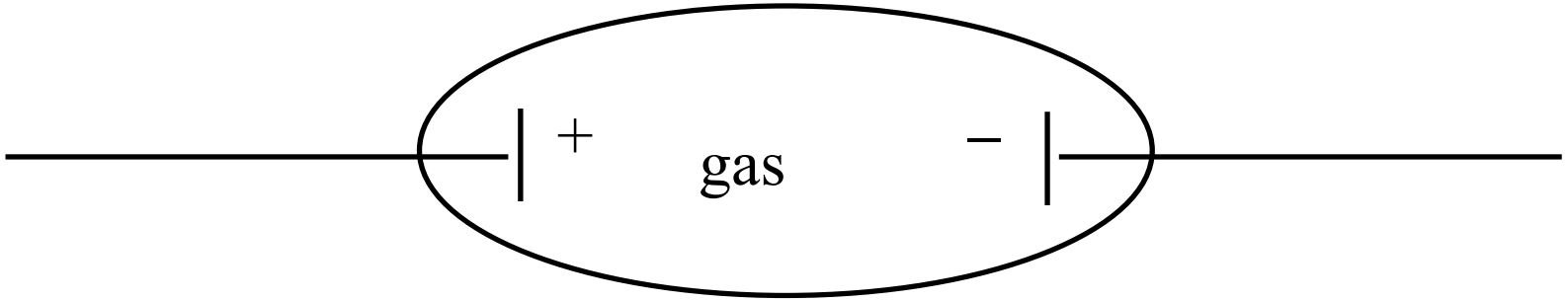


(*a*)

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Continuous Spectrum

- Continuous unbroken spectrum of all colors
  - i.e., visible light through a prism
- Consider light given off when spark passes through gas under vacuum



- Spark (electrical discharge) excites gas molecules (atoms)

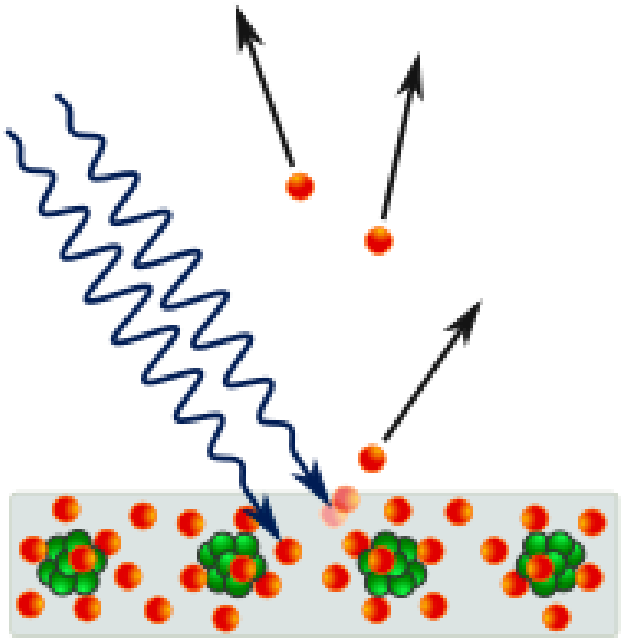
Light as Particles

- Max Planck and Albert Einstein (1905)
  - Electromagnetic radiation is stream of small packets of energy
  - **Quanta of energy** or **photons**
  - Each photon travels with **velocity = c**
  - Pulses with frequency = **ν**

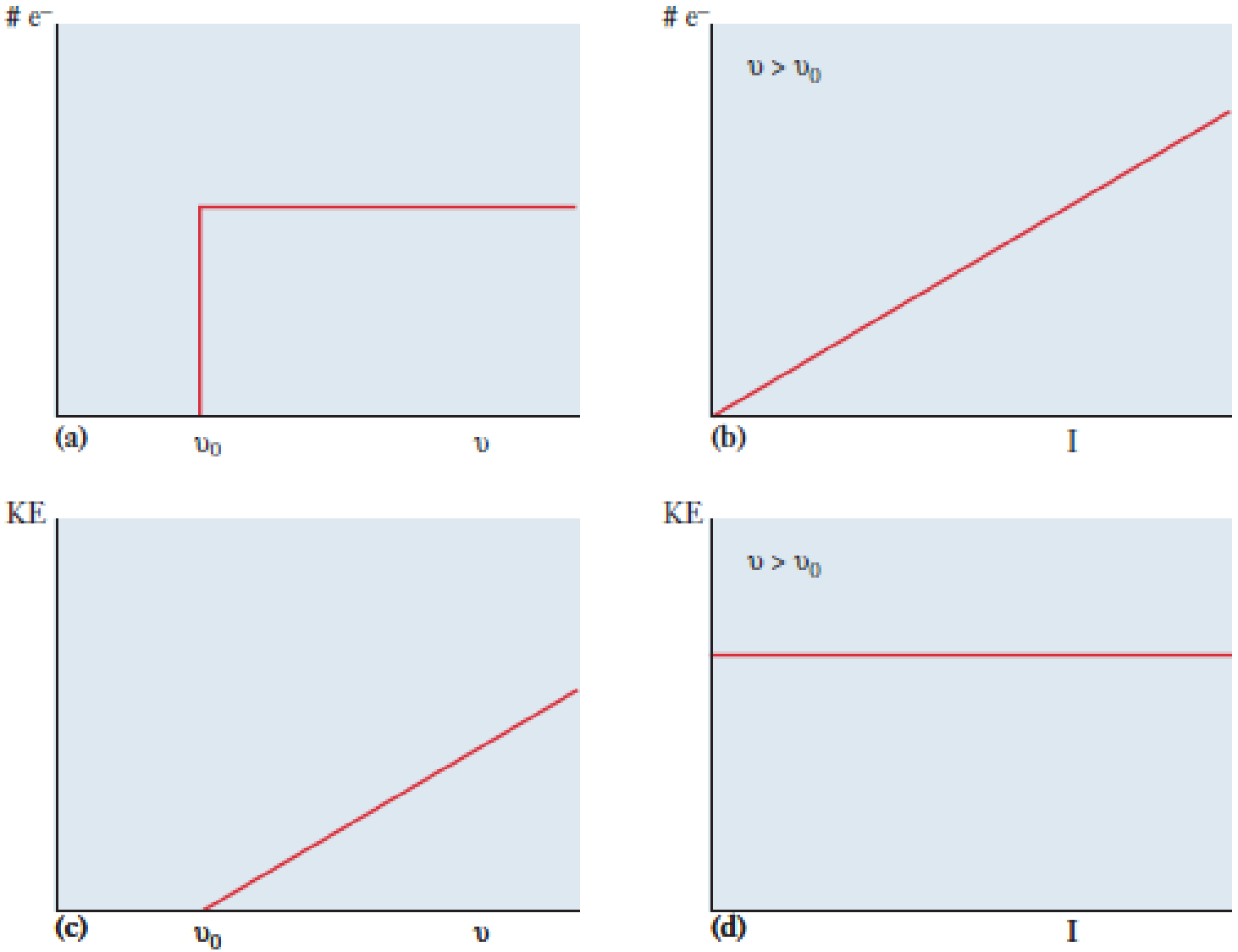
- Energy of photon of electromagnetic radiation is proportional to its frequency
  - Energy of photon $\mathbf{E = h\nu}$
  - ***h*** = Planck's constant
    = **6.626×10^{-34} J·s**

Photoelectric Effect

Experiments that brought Einstein with a Nobel Prize

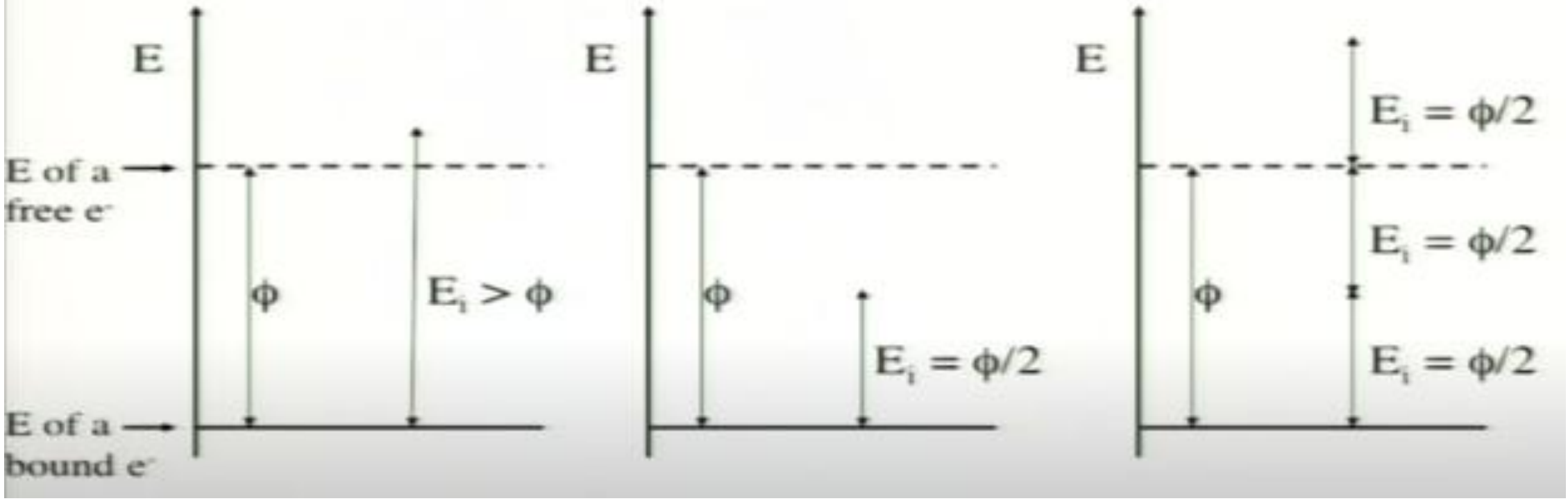


Photoelectric Effect Experimental Results

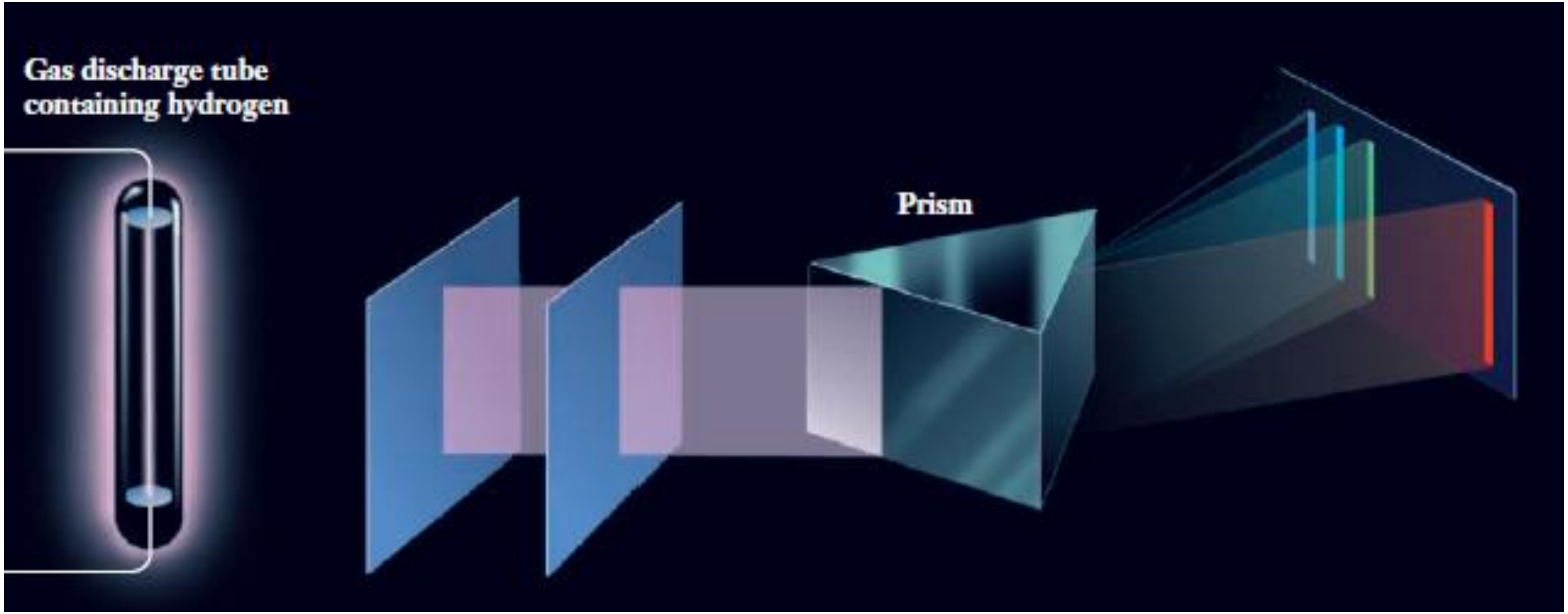


Exercise

The energy of an incoming photon must be equal to or greater than the ϕ of the metal in order to eject an electron.

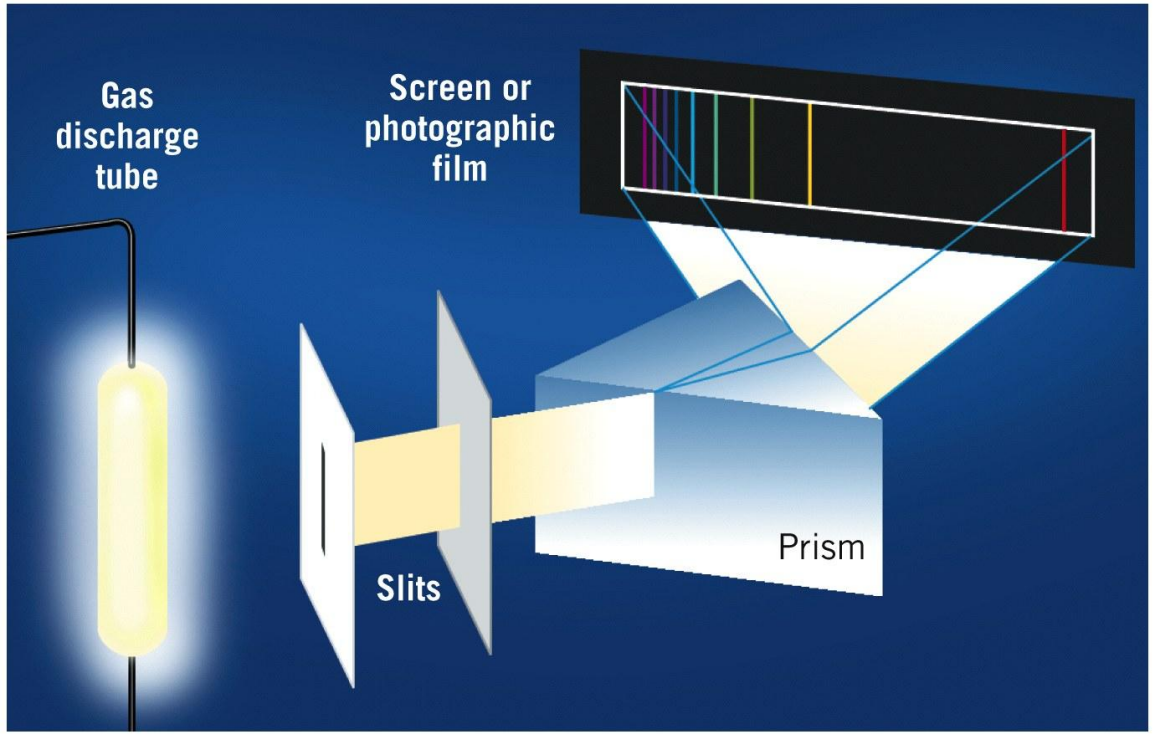


Hydrogen Line Spectrum



Line Spectrum

- Spectrum that has only a few discrete lines
- Also called **atomic spectrum** or **emission spectrum**
- Each element has unique emission spectrum



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Atomic Spectra

- Atomic line spectra are rather complicated
- Line spectrum of hydrogen is simplest
  - Single electron
  - 1st success in explaining quantized line spectra
  - 1st studied extensively
- J.J. Balmer
  - Found empirical equation to fit lines in visible region of spectrum
- J. Rydberg
  - More general equation explains all emission lines in H atom spectrum (IR, Vis, and UV)

Rydberg Equation

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

R_H = 109,678 cm^{-1} = Rydberg constant

λ = wavelength of light emitted

n_1 & n_2 = whole numbers (integers) from 1 to ∞ where $n_2 > n_1$

If n_1 = 1, then n_2 = 2, 3, 4, …

} Corresponds to allowed energy levels for atom

- Can be used to calculate all spectral lines of hydrogen

Exercise

Consider the Balmer series where $n_1 = 2$. Calculate λ (in *nm*) for the transition from $n_2 = 6$ down to $n_1 = 2$.

$$\frac{1}{\lambda} = R_H\left(\frac{1}{2^2} - \frac{1}{6^2}\right) = 109{,}678\text{cm}^{-1}\left(\frac{1}{4} - \frac{1}{36}\right) = 24{,}373\text{cm}^{-1}$$

$$\lambda = \frac{1}{24{,}372.9\text{cm}^{-1}} = 4.1029 \times 10^{-5}\text{cm} \times \frac{1\text{m}}{100\text{cm}} \times \frac{1\text{nm}}{1 \times 10^{-9}\text{m}}$$

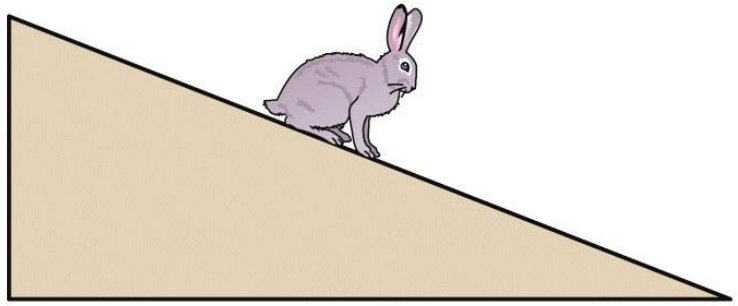
λ = 410.3 nm Violet line in spectrum

Significance of Atomic Spectra

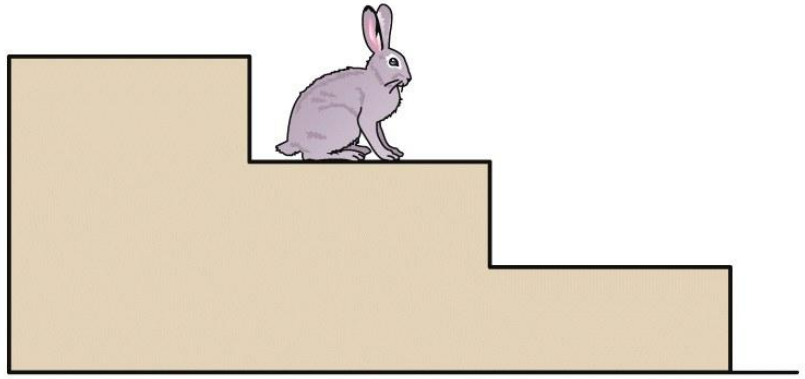
- Atomic line spectra tells us
  - When excited atom loses energy
    - Only fixed amounts of energy can be lost
    - Only certain **energy** photons are emitted
    - Electron restricted to certain fixed **energy levels** in atoms
  - Energy of electron is **quantized**
  - Simple extension of Planck's Theory
- Any theory of atomic structure must account for
  - Atomic spectra
  - Quantization of energy levels in atom

What Does Quantized Mean ?

Potential Energy of Rabbit



(*a*) Any potential energy allowed: energy values are *continuous*



(*b*) Potential energy restricted: energy values are *discrete*

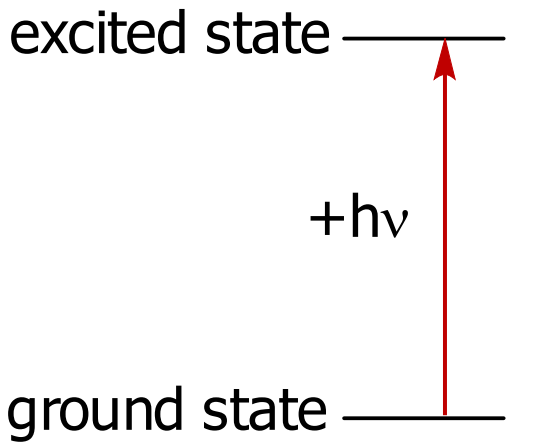
- Energy is quantized if only certain discrete values are allowed
- Presence of discontinuities makes atomic emission quantized

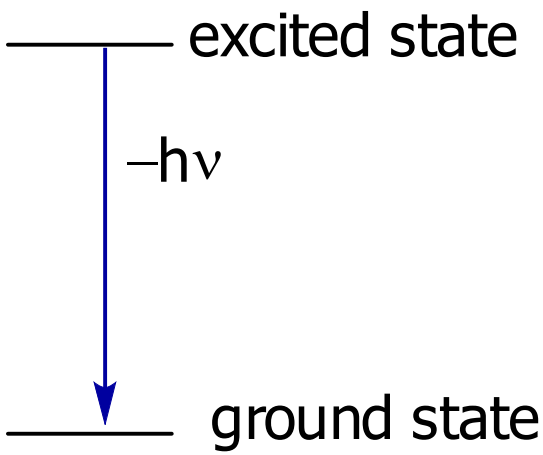
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Electronic Structure of Atom

Most information comes from:

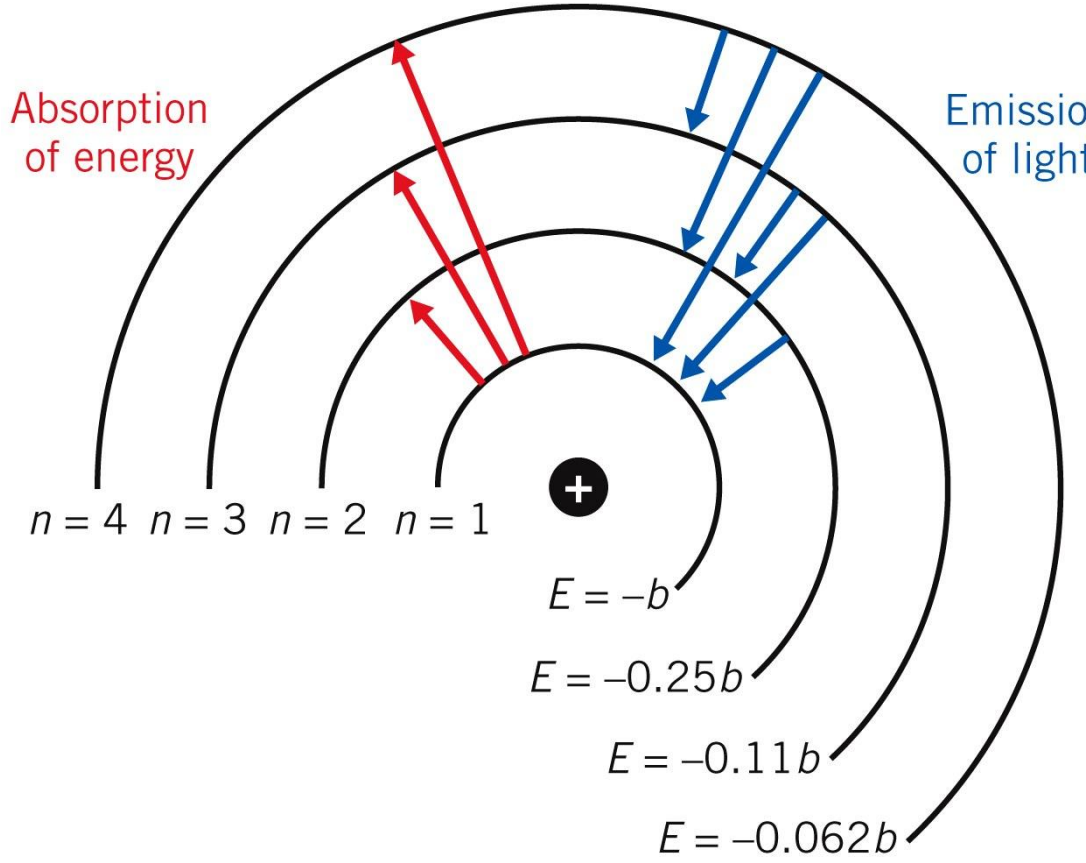
1. Study of light absorption
   - Electron absorbs energy
   - Moves to higher energy "excited state"
2. Study of light emission
   - e^- loses photon of light
   - Drops back down to lower energy "ground state"





Energy Level Diagram for Hydrogen Atom

- Absorption of photon
  - Electron raised to higher energy level
- Emission of photon
  - Electron falls to lower energy level

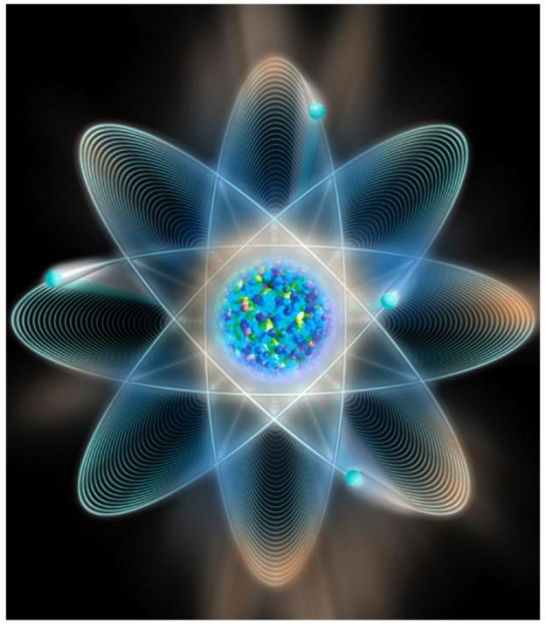


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- ***E*** s are quantized
  - Every time e^- drops from ***n*** = 3 to ***n*** = 1
  - Same frequency photon is emitted
  - Yields line spectra

Bohr Model of Atom

- 1st theoretical model of atom to successfully account for Rydberg equation
  - Quantization of Energy in Hydrogen atom
  - Correctly explained Atomic Line Spectra
- Proposed that electrons moved around nucleus like planets move around sun
  - Move in fixed paths or orbits
  - Each orbit has fixed energy



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Energy for Bohr Model of Hydrogen

$$E \propto -\frac{1}{n^2} \qquad b = \frac{2\pi^2 me^4}{h^2}$$

- Equation for energy of e^- in H atom

  - Ultimately ***b*** relates to R_H by $b = R_H hc$
  - OR

$$E = -\frac{b}{n^2} = -\frac{R_H hc}{n^2}$$

  - where $b = R_H hc = 2.1788 \times 10^{-18}$ *J/atom*
  - Allowed values of ***n*** = 1, 2, 3, 4, …∞
  - ***n*** = Quantum number
  - Used to identify orbit

Bohr Model of Hydrogen

- ***E*** is negative number
- Reference point ***E*** = 0 when ***n*** = ∞
  - e^- not attached to nucleus
  - Sign arises from Coulombic attraction between + and – charges (oppositely charged bodies)
- Coulomb's Law Attractive force

$$E = -\frac{(\text{charge on A})(\text{charge on B})}{\text{distance between them}}$$

- Stronger attractive force = more negative ***E***

Bohr Model of Hydrogen

- ***n*** = 1 1st Bohr orbit
  - Most stable ***E*** state = ground state = Lowest ***E*** state
  - Electron remains in lowest ***E*** state unless disturbed

How to disturb the atom?

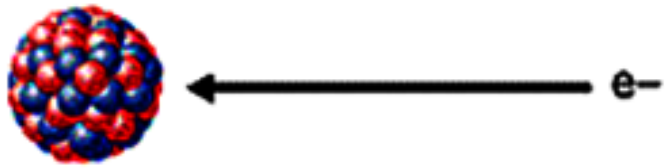
- Add $E = h\nu$
- e^- raised higher ***n*** orbit ***n*** = 2, 3, 4, … ∞
- Higher ***n*** orbits = **excited states = less stable**
- **So** e^- quickly drops to lower *E* orbit and emits photon of *E* equal to ΔE between levels

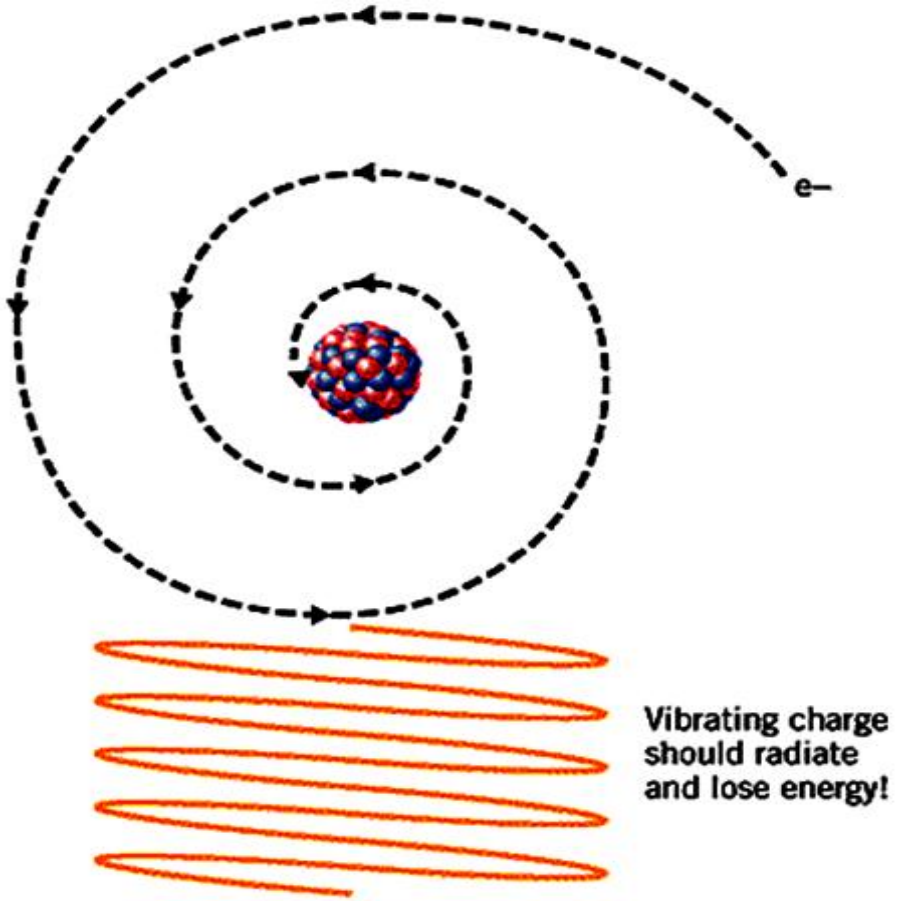
$\Delta E = E_h - E_l$ ***h*** **= higher** ***l*** **= lower**

Bohr's Model Fails

- Theory could not explain spectra of multi- electron atoms
- Theory doesn't explain collapsing atom paradox
- If e^- doesn't move, atom collapses

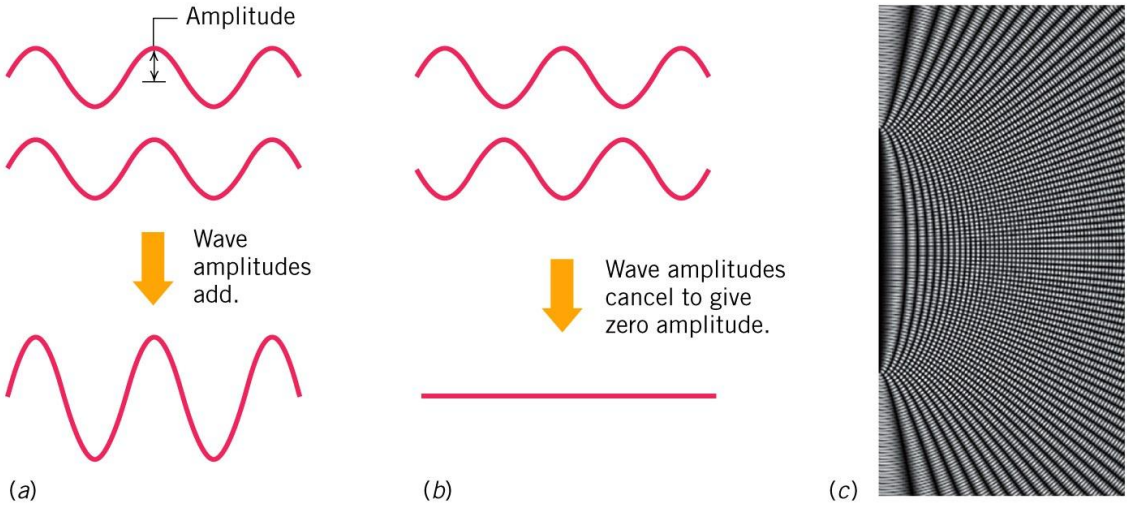
- Positive nucleus should easily capture e^-
- Vibrating charge should radiate and lose energy





Diffraction and Electrons

- Light
  - Exhibits interference
  - Has particle nature
- Electrons
  - Known to be particles
  - Also demonstrate interference



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Schrödinger Equation



$$\hat{H}\Psi = E \cdot \Psi$$

Ψ = wavefunction

E = binding energy

$\hat{H}$ = Hamiltonian operator

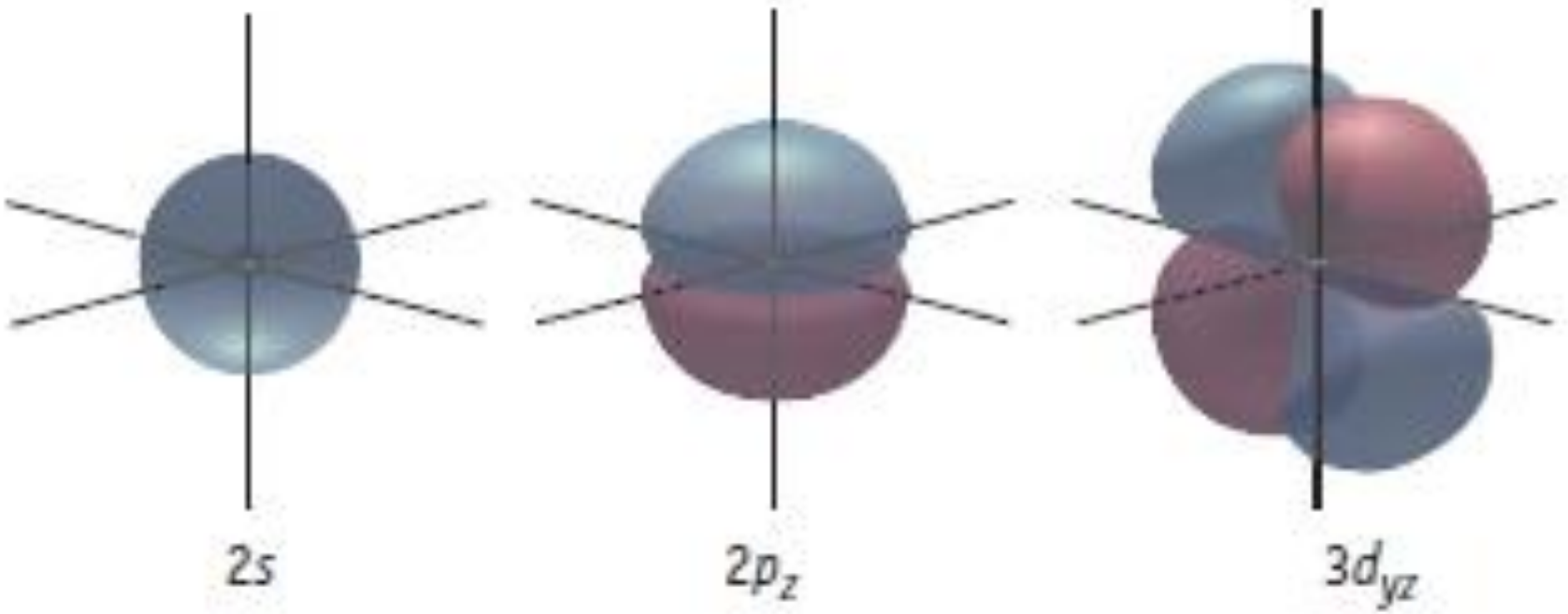
Wave Functions and Atomic Orbitals

Orbital	Wave Function
$1s$	$\psi_{1s} = \pi^{-1/2} a_0^{-3/2} e^{-(r/a_0)}$
$2s$	$\psi_{2s} = (4\pi)^{-1/2}(2a_0)^{-3/2}\left(2 - \frac{r}{a_0}\right) e^{-(r/2a_0)}$
$2p_x$	$\psi_{2p_x} = (4\pi)^{-1/2}(2a_0)^{-3/2} \frac{r}{a_0} e^{-(r/2a_0)} \sin\theta \cos\phi$
$2p_y$	$\psi_{2p_y} = (4\pi)^{-1/2}(2a_0)^{-3/2} \frac{r}{a_0} e^{-(r/2a_0)} \sin\theta \sin\phi$
$2p_z$	$\psi_{2p_z} = (4\pi)^{-1/2}(2a_0)^{-3/2} \frac{r}{a_0} e^{-(r/2a_0)} \cos\theta$

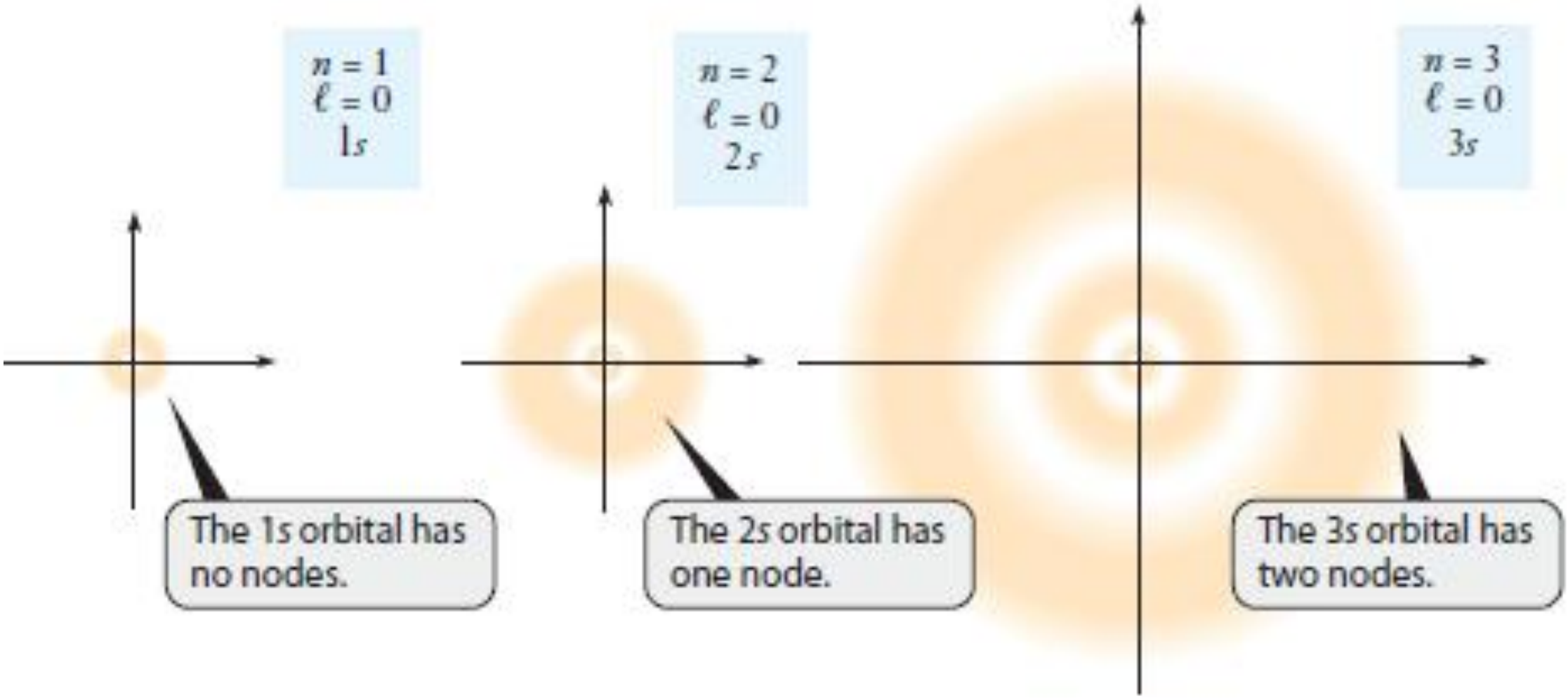
Quantum and Magnetic Number

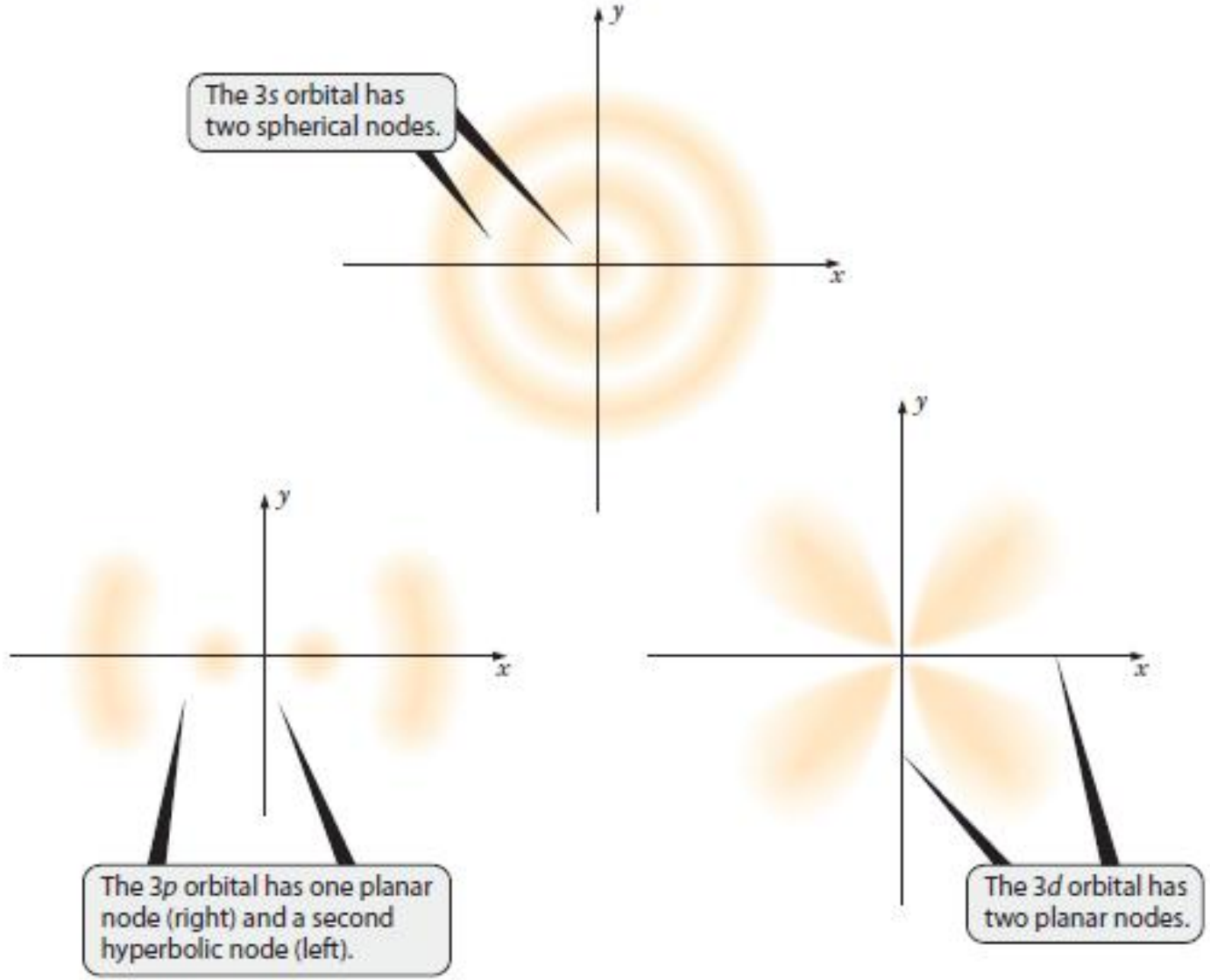
Value of n	Values for ℓ (letter designation)	Values for m_ℓ	Number of Orbitals
1	0 (s)	0	1
2	0 (s)	0	1
	1 (p)	$-1, 0, 1$	3
3	0 (s)	0	1
	1 (p)	$-1, 0, 1$	3
	2 (d)	$-2, -1, 0, 1, 2$	5
4	0 (s)	0	1
	1 (p)	$-1, 0, 1$	3
	2 (d)	$-2, -1, 0, 1, 2$	5
	3 (f)	$-3, -2, -1, 0, 1, 2, 3$	7

Atomic Orbital Shapes

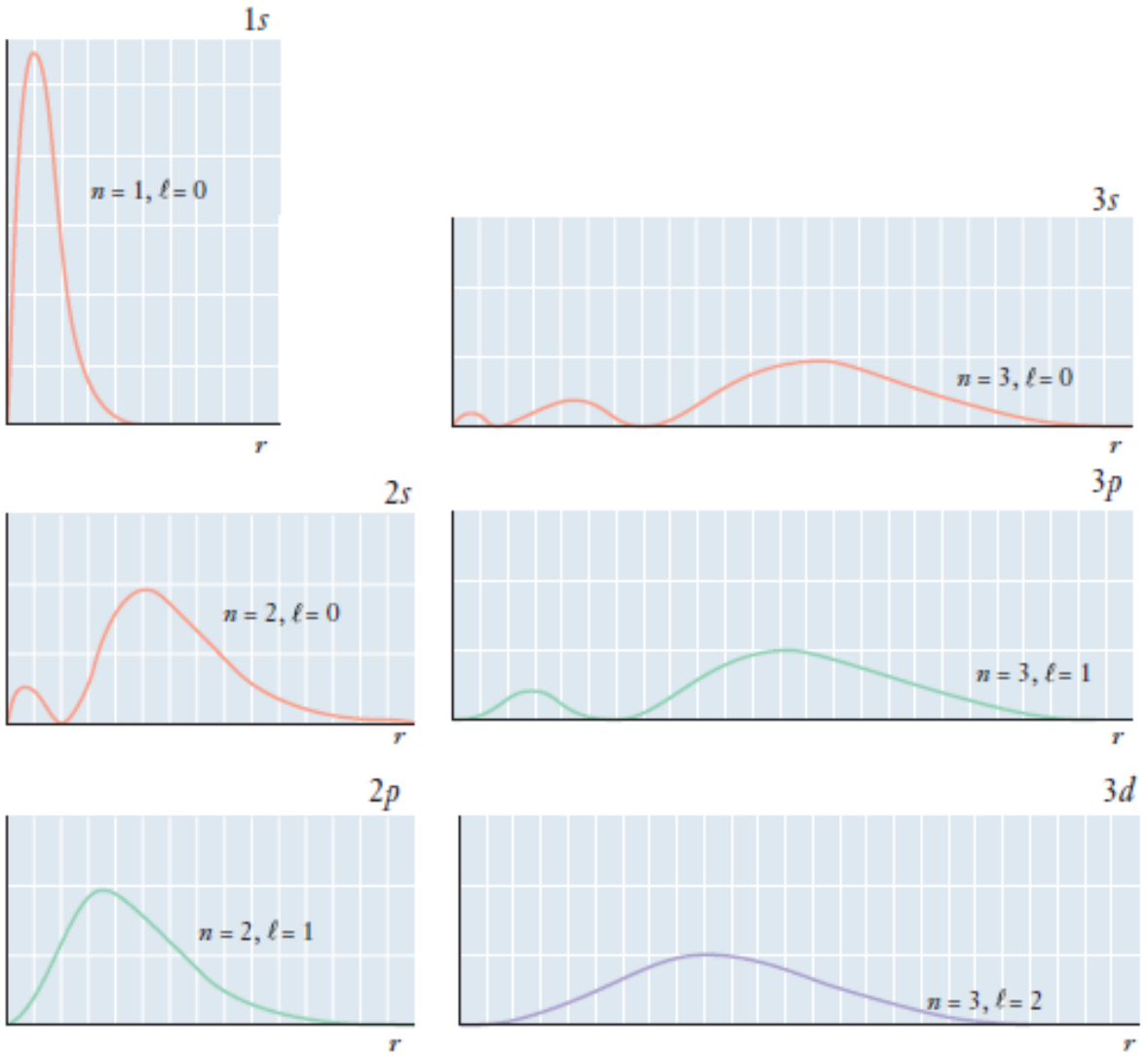


Atomic Orbital Shapes

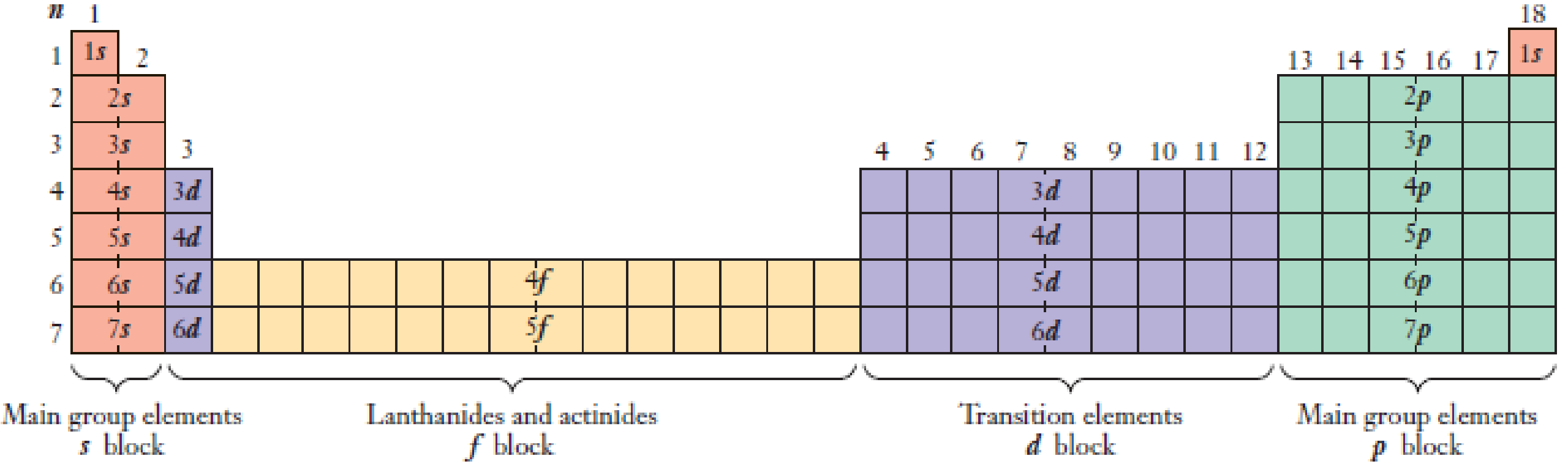



y
x
The 3s orbital has two spherical nodes.
y
x
The 3p orbital has one planar node (right) and a second hyperbolic node (left).
y
x
The 3d orbital has two planar nodes.

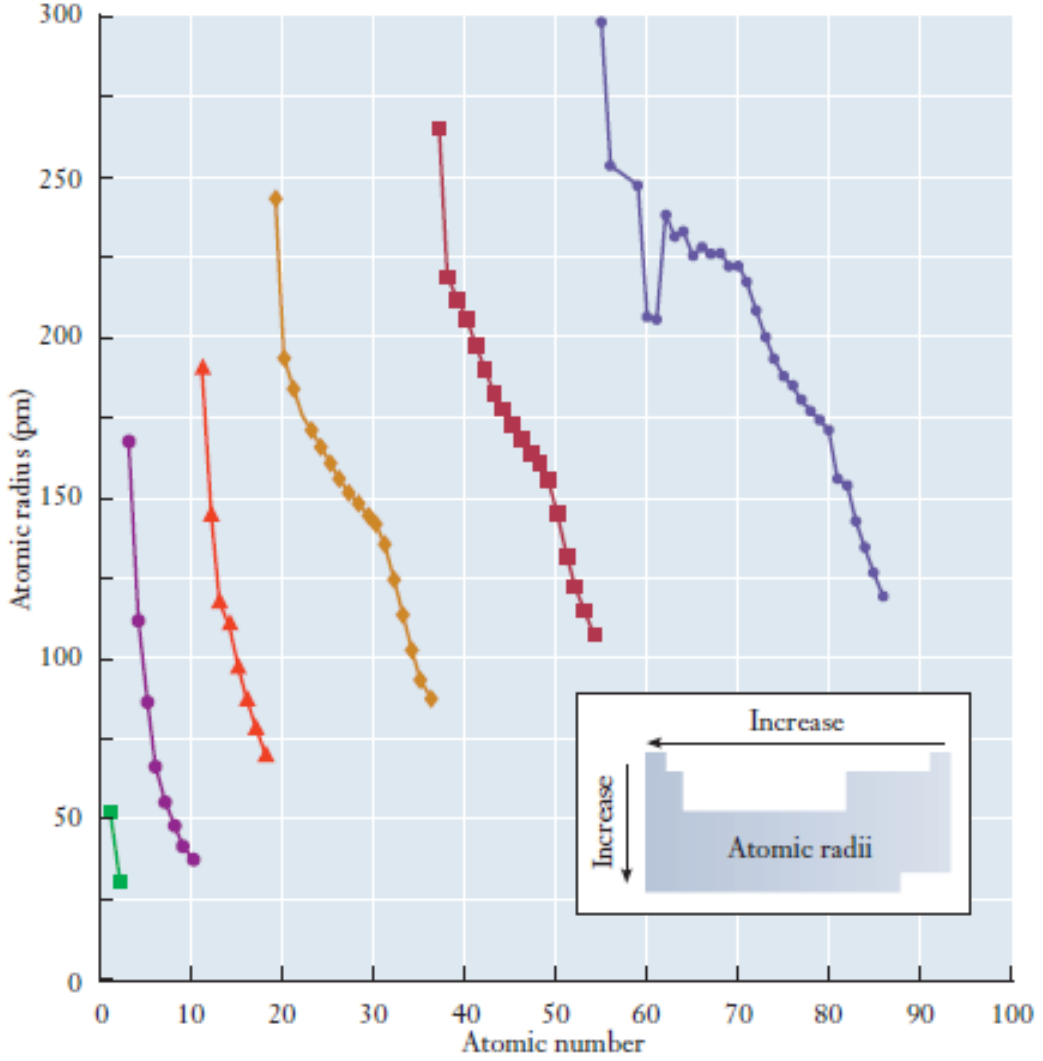
Electron Probability Distribution in Atomic Orbitals



Classification of Orbital in the Periodic Table



Relationship between Atomic Radii and Atomic Number



First Ionization Energy for Elements

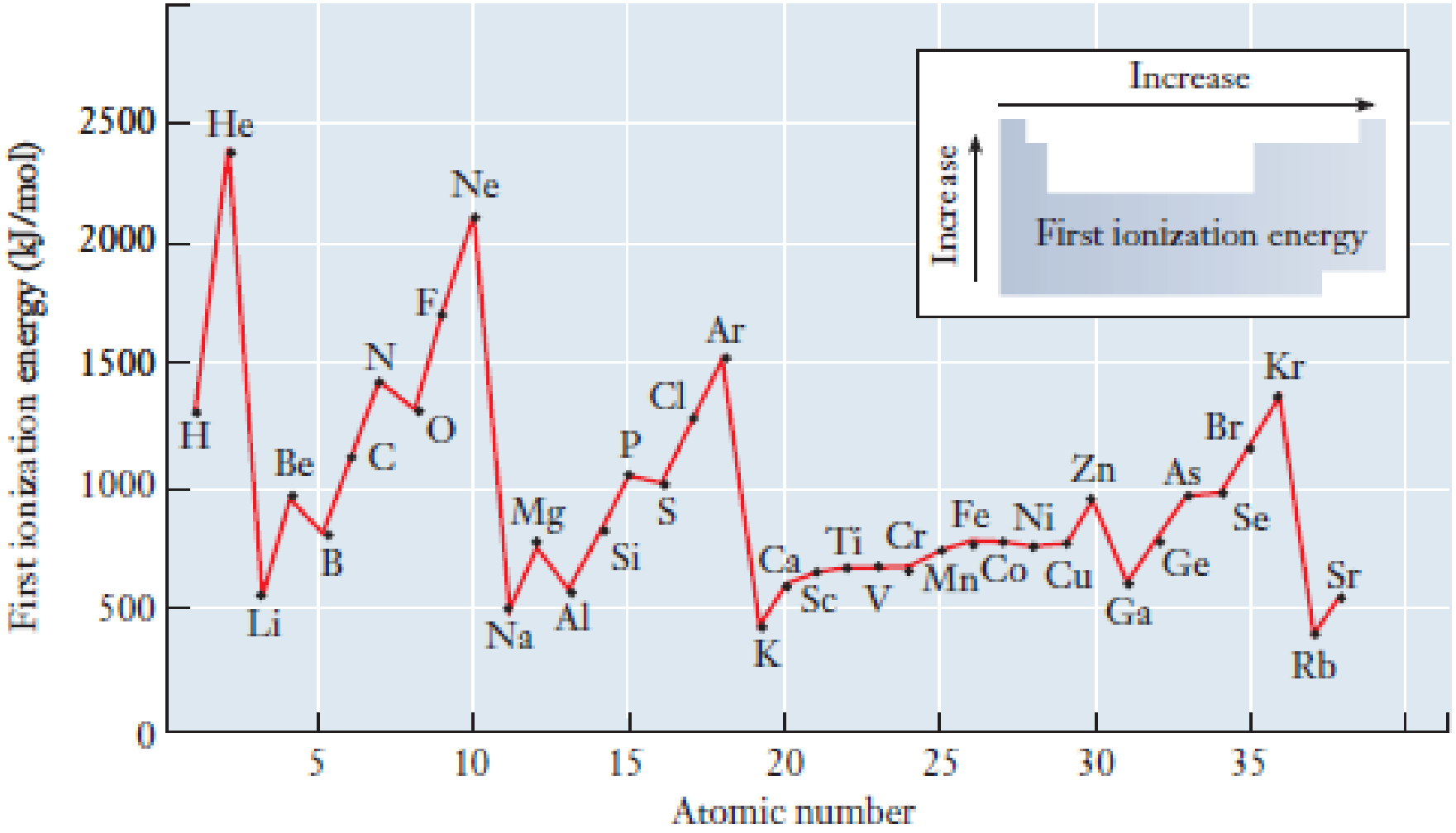


Table of Ionization Energies for Elements

Z	Element	IE_1	IE_2	IE_3	IE_4
1	H	1,312	—	—	—
2	Hc	2,372	5,250	—	—
3	Li	520.2	7,298	11,815	—
4	Bc	899.4	1,757	14,848	21,007
5	B	800.6	2,427	3,660	25,026
6	C	1,086	2,353	4,620	6,223
7	N	1,402	2,856	4,578	7,475
8	O	1,314	3,388	5,300	7,469
9	F	1,681	3,374	6,050	8,408
10	Nc	2,081	3,952	6,122	9,370
11	Na	495.6	4,562	6,912	9,544
12	Mg	737.7	1,451	7,733	10,540
13	Al	577.6	1,817	2,745	11,578
14	Si	786.4	1,577	3,232	4,356
15	P	1,012	1,908	2,912	4,957
16	S	999.6	2,251	3,357	4,564
17	Cl	1,251	2,297	3,822	5,158
18	Ar	1,520	2,666	3,931	5,771

Electron Affinity in Elements

